Mathematical Analysis

AshaIy

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Version 0.1.0

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Source Code & Contributions
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Feel free to open issues or pull requests for corrections.

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Contents

Preface v

1 Basic Topology 1
- 1.1 Metric Spaces . . . 1
- 1.2 Compact Sets . . . 1
- 1.3 Connected Sets . . . 1

2 Sequences and Series 2
- 2.1 Convergent Sequences . . . 2
- 2.2 Subsequences . . . 2
- 2.3 Cauchy Sequences . . . 2
- 2.4 Upper and Lower Limits . . . 2
- 2.5 Some Special Sequences . . . 2
- 2.6 Series . . . 2
- 2.7 Series of Nonnegative Terms . . . 2
- 2.8 The Number e . . . 2
- 2.9 The Root and Ratio Tests . . . 2
- 2.10 Power Series . . . 2
- 2.11 Summation by Parts . . . 2
- 2.12 Absolute Convergence . . . 2
- 2.13 Addition and Multiplication of Series . . . 3
- 2.14 Rearrangements . . . 3

3 Continuity 4
- 3.1 Limits of Functions . . . 4
- 3.2 Continuous Functions . . . 4
- 3.3 Continuity and Compactness . . . 4
- 3.4 Continuity and Connectedness . . . 4
- 3.5 Discontinuities . . . 4
- 3.6 Monotonic Functions . . . 4
- 3.7 Infinite Limits and Limits at Infinity . . . 4

4 Differentiation **5**
4.1 The Derivative of a Real Function . . . 5
4.2 Mean Value Theorems . . . 5
4.3 The Continuity of Derivatives . . . 5
4.4 L'Hospital's Rule . . . 5
4.5 Derivatives of Higher Order . . . 5
4.6 Taylor's Theorem . . . 5
4.7 Differentiation of Vector-valued Functions . . . 5

5 The Riemann-Stieltjes Integral **6**
5.1 Definition and Existence of the Integral . . . 6
5.2 Properties of the Integral . . . 6
5.3 Integration and Differentiation . . . 6
5.4 Integration of Vector-valued Functions . . . 6
5.5 Rectifiable Curves . . . 6

6 Sequences and Series of Functions **7**
6.1 Discussion of Main Problem . . . 7
6.2 Uniform Convergence . . . 7
6.3 Uniform Convergence and Continuity . . . 7
6.4 Uniform Convergence and Integration . . . 7
6.5 Uniform Convergence and Differentiation . . . 7
6.6 Equicontinuous Families of Functions . . . 7
6.7 The Stone-Weierstrass Theorem . . . 7

7 Some Special Functions **8**
7.1 Power Series . . . 8
7.2 The Exponential and Logarithmic Functions . . . 8
7.3 The Trigonometric Functions . . . 8
7.4 The Algebraic Completeness of the Complex Field . . . 8
7.5 Fourier Series . . . 8
7.6 The Gamma Function . . . 8

8 Functions of Several Variables **9**
8.1 Differentiation . . . 9
8.2 The Contraction Principle . . . 9
8.3 The Inverse Function Theorem . . . 9
8.4 The Implicit Function Theorem . . . 9
8.5 The Rank Theorem . . . 9
8.6 Derivatives of Higher Order . . . 9
8.7 Differentiation of Integrals . . . 9

9 Integration of Differential Forms **10**
9.1 Integration . . . 10
9.2 Primitive Mappings . . . 10
9.3 Partitions of Unity . . . 10
9.4 Change of Variables . . . 10
9.5 Differential Forms . . . 10
9.6 Simplexes and Chains . . . 10
9.7 Stokes' Theorem . . . 10
9.8 Closed Forms and Exact Forms . . . 10
9.9 Vector Analysis . . . 10

10 The Lebesgue Theory **11**
10.1 Set Functions . . . 11
10.2 Construction of the Lebesgue Measure . . . 11
10.3 Measure Spaces . . . 11
10.4 Measurable Functions . . . 11
10.5 Simple Functions . . . 11
10.6 Integration . . . 11
10.7 Comparison with the Riemann Integral . . . 11
10.8 Integration of Complex Functions . . . 11
10.9 Functions of Class $\mathscr{L}^2$ . . . 11

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.34.0 and Mathlib v4.34.0 (revision `5ed296525643`).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.

Chapter 1

Basic Topology

1.1 Metric Spaces

1.2 Compact Sets

1.3 Connected Sets

Chapter 2

Sequences and Series

2.1 Convergent Sequences

2.2 Subsequences

2.3 Cauchy Sequences

2.4 Upper and Lower Limits

2.5 Some Special Sequences

2.6 Series

2.7 Series of Nonnegative Terms

2.8 The Number e

2.9 The Root and Ratio Tests

2.10 Power Series

2.11 Summation by Parts

2.12 Absolute Convergence

2.13 Addition and Multiplication of Series

2.14 Rearrangements

Chapter 3

Continuity

3.1 Limits of Functions

3.2 Continuous Functions

3.3 Continuity and Compactness

3.4 Continuity and Connectedness

3.5 Discontinuities

3.6 Monotonic Functions

3.7 Infinite Limits and Limits at Infinity

Chapter 4

Differentiation

4.1 The Derivative of a Real Function

4.2 Mean Value Theorems

4.3 The Continuity of Derivatives

4.4 L'Hospital's Rule

4.5 Derivatives of Higher Order

4.6 Taylor's Theorem

4.7 Differentiation of Vector-valued Functions

Chapter 5

The Riemann-Stieltjes Integral

5.1 Definition and Existence of the Integral

5.2 Properties of the Integral

5.3 Integration and Differentiation

5.4 Integration of Vector-valued Functions

5.5 Rectifiable Curves

Chapter 6

Sequences and Series of Functions

6.1 Discussion of Main Problem

6.2 Uniform Convergence

6.3 Uniform Convergence and Continuity

6.4 Uniform Convergence and Integration

6.5 Uniform Convergence and Differentiation

6.6 Equicontinuous Families of Functions

6.7 The Stone-Weierstrass Theorem

Chapter 7

Some Special Functions

7.1 Power Series

7.2 The Exponential and Logarithmic Functions

7.3 The Trigonometric Functions

7.4 The Algebraic Completeness of the Complex Field

7.5 Fourier Series

7.6 The Gamma Function

Chapter 8

Functions of Several Variables

8.1 Differentiation

8.2 The Contraction Principle

8.3 The Inverse Function Theorem

8.4 The Implicit Function Theorem

8.5 The Rank Theorem

8.6 Derivatives of Higher Order

8.7 Differentiation of Integrals

Chapter 9

Integration of Differential Forms

9.1 Integration

9.2 Primitive Mappings

9.3 Partitions of Unity

9.4 Change of Variables

9.5 Differential Forms

9.6 Simplexes and Chains

9.7 Stokes' Theorem

9.8 Closed Forms and Exact Forms

9.9 Vector Analysis

Chapter 10

The Lebesgue Theory

10.1 Set Functions

10.2 Construction of the Lebesgue Measure

10.3 Measure Spaces

10.4 Measurable Functions

10.5 Simple Functions

10.6 Integration

10.7 Comparison with the Riemann Integral

10.8 Integration of Complex Functions

10.9 Functions of Class $\mathscr{L}^2$

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